

Quiz 5

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Which of the following subsets of $\mathbf{R}^3$ are actually subspaces? (no explanation needed)

(a) The plane of vectors (b_1, b_2, b_3) with $b_1 = b_2$.

Yes

(b) The plane of vectors (b_1, b_2, b_3) with $b_1 = 1$.

No

(c) All linear combinations of $\mathbf{v} = (1, 4, 0)$ and $\mathbf{w} = (2, 2, 2)$.

Yes

(d) All vectors that satisfy $b_1 + b_2 + b_3 = 0$.

Yes

(e) All vectors that satisfy $b_1 \leq b_2 \leq b_3$.

No

(f) All vectors (b_1, b_2, b_3) that are not in the column space $C(\mathbf{A})$ where $\mathbf{A}$ is an 3×3 matrix.

No

Problem 2. (4 points) Describe the column space of $\mathbf{A}$, $C(\mathbf{A})$, where $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 0 & 0 \end{bmatrix}$.

$$\mathbf{A}\mathbf{x} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \begin{bmatrix} x_1 + 2x_2 \\ x_1 + 2x_2 \\ 0 \end{bmatrix}$$

$$= (x_1 + 2x_2) \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$= c \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$\Rightarrow C(\mathbf{A}) = \left\{ c \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \mid c \text{ is any real number} \right\}$$